

Case Study - SW Testing Engineer

Session Management — EVSE Charger Firmware

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Agenda

1. Building a test basis without formal requirements
2. Exercise 1 — Black-box test case design
3. Exercise 2 — Test case with AI assistance
4. Q&A

Building a test basis without formal requirements

Sources and confidence

Source	Confidence	Use
Standards	High (interfaces, safety)	Invariants, external sequences
Safety/physics invariants	High	“Never” rules
Developers	Medium (intent, biased toward what they built)	Rationale
Code	High for does, zero for should	Model extraction, not an oracle
Field logs, tickets	High relevance, medium correctness	Prioritization
Old docs, existing tests	Low	Leads only
Previous release	Consistency only	Regression testing

Behaviour: bug or intended

Behaviour	Conclusion
Violates a standard or safety requirement	BUG
Harmful for user or operator	BUG ¹
Contradicts developer intent	BUG
Intended and working	INTENDED
Nobody can decide	PROVISIONAL

Behaviour: bug or intended

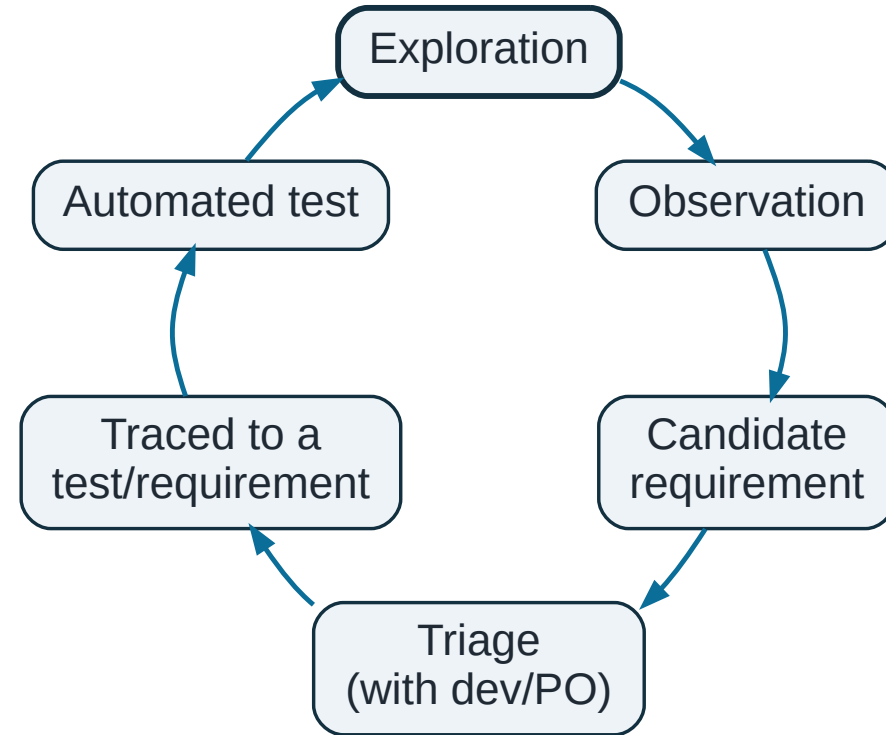
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Decide, document, and compound

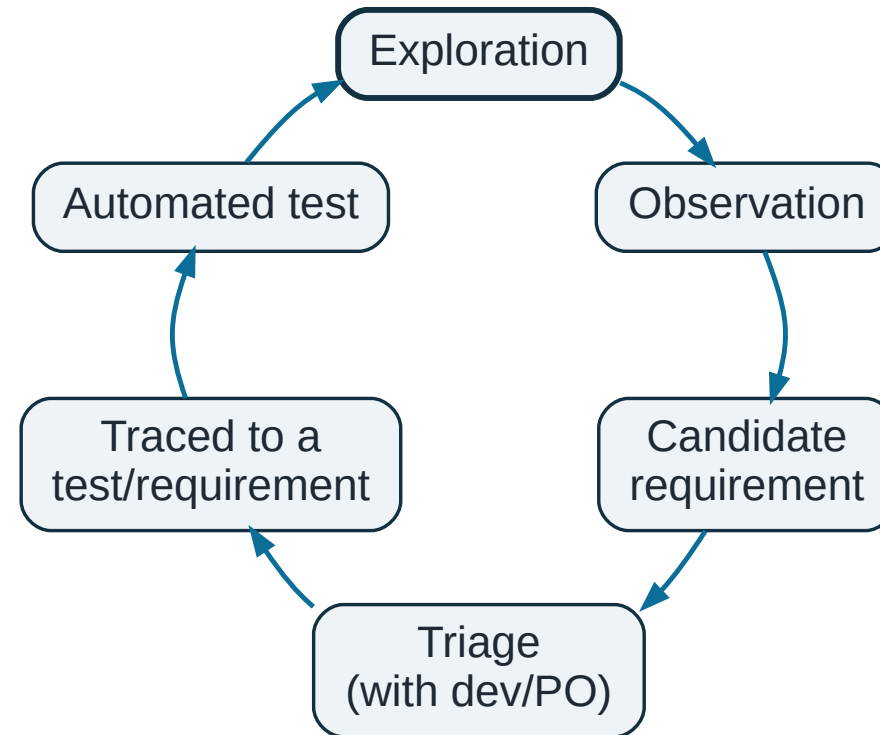
Use Behaviour Decision Records to track decisions that compound into specifications. Each BDR records:

- ID
- Observed behavior
- FW version
- Decision
- Rationale
- Who decided
- Linked requirement/test IDs
- Sources consulted

The workflow..



The workflow..



Possible health metrics:

- % of confirmed cells in the **state** x **event** matrix
- % of requirements still “observed”, tracked per release

..prevents freezing incorrect behaviour

Two test classes:

- **Requirement tests:** based on confirmed behaviour (a failure triggers a bug)
- **Provisional tests:** based on observed behaviour (a failure triggers a review)

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- **Provisional tests:** based on observed behaviour (a failure triggers a review)

	Requirement test	Provisional test
Expected result from	Baselined requirement	Current observed behavior or best hypothesis
Fail means	Bug	Behavior changed → triage (bug or decision)
Pass means	Validation evidence	Nothing changed
Counts for release sign-off	Yes	No, reported separately*
Linked to	Requirement ID	Open question ID + owner + deadline**

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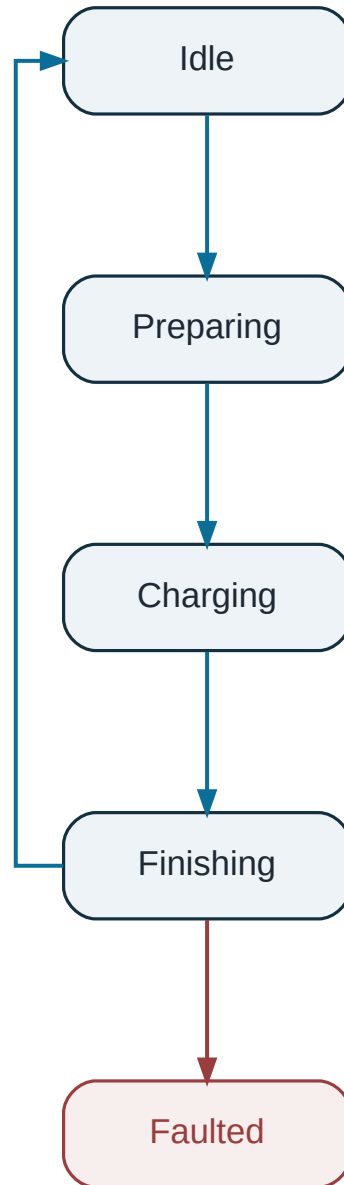
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Pass means	Validation evidence	Nothing changed
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Linked to	Requirement ID	Open question ID + owner + deadline**

* Provisional tests still run, but they are not blocking.

** Provisional tests **must expire**. If they sit unresolved for several releases, they silently become frozen incorrect behaviour.

Exercise 1

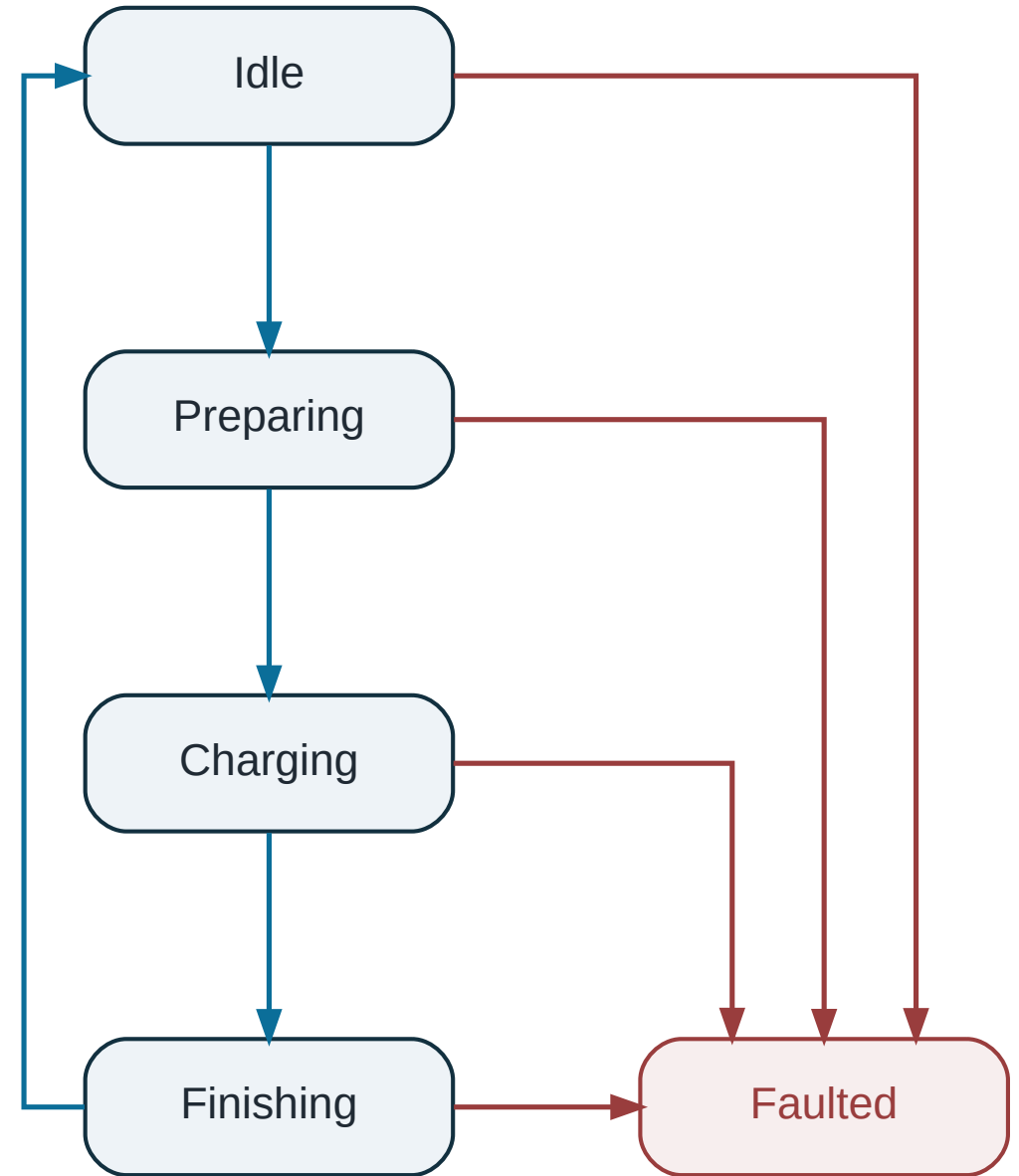
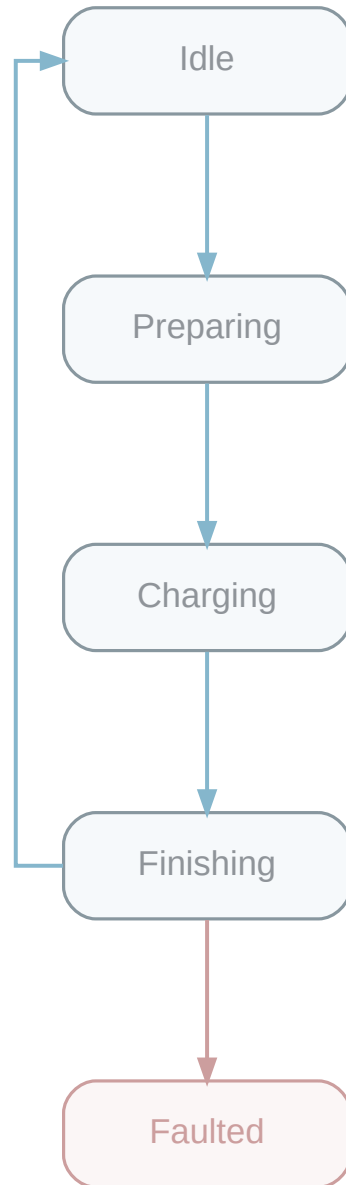
Interpretation of the state machine



State Machine is ambiguous and partial (as per OCPP):

- Faulted should be reachable from all states.
- Missing states (Suspended, Unavailable, ...)

Interpretation of the state machine



Test cases

“Max limit allowed by the connector” is ambiguous: **[assumption]** the effective limit is the minimum of EVSE, EV, cable, ...

* every expected result needs a value, a tolerance, and a time limit

** and a large value, e.g. I_{max}

Test cases

“Max limit allowed by the connector” is ambiguous: [assumption] the effective limit is the minimum of EVSE, EV, cable, ...

Preconditions:

- Mocks / HIL setup to simulate an EV and a DC load
- Connector limit known ([assumption] $I_{max} = 200\text{ A}$)
- The charger is in Idle
- No active faults

ID	Type	Input	Expected result *
TC-01	Nominal	EV requests exactly I_{max}	Idle→Preparing→Charging. Measured $I = I_{max} \pm \text{tolerance}$, with no overshoot beyond tolerance during ramp-up. Stable for N minutes, no fault. Meter values are consistent
TC-02	Boundary (above)	EV requests $I_{max} + 1\text{ A}$ **	Charger advertises $\text{EVSEMaxCurrent} = I_{max}$ and delivers $\leq I_{max}$. The session is not rejected or faulted, just limited
TC-03	Boundary (below)	EV requests $I_{max} - 1\text{ A}$	Delivers the requested value exactly. This proves the limiting isn't over-aggressive or off-by-one
TC-04	Error	While charging at I_{max} , inject a power module fault	Transition to Faulted, current ramps to 0 A, contactors open within the specified time, session closed with the correct final meter value

* every expected result needs a value, a tolerance, and a time limit

** and a large value, e.g. I_{max}

Exercise 1: Test Case TC-01

Test Case ID: TC-SM-IMAX-001 (TC-01)

Title: Start charging session with EV requesting exactly the connector maximum current

Priority: High

Objective: Verify that when the EV requests a current equal to I_{max} , the charger goes through Idle → Preparing → Charging, delivers I_{max} within tolerance without overshoot, keeps it stable, and meters the energy correctly.

Preconditions

- Firmware version recorded.
- The charger is powered on and in Idle state.
- The charger is connected to the CSMS simulator.
- An EV emulator is connected.
- A power analyzer is on the DC output.
- I_{max} is configured, and its value is recorded from the charger configuration.
- No active faults are present.

Test Data

Parameter	Value
Output voltage	400 V DC
EV requested current	$I_{max} = 400\text{ A}$
Hold time	T_{stable}

Test Steps and Expected Results

Step	Action	Expected Result
1	EV emulator runs charge parameter discovery with EV max current = I_{max}	Charger advertises EVSE max current $\geq I_{max}$ (checked in the protocol log).
2	Let cable check and precharge complete	Both complete without error. No isolation fault.
3	EV emulator requests target current = I_{max}	State Preparing → Charging.
4	Observe the current ramp-up on the power analyzer	Current reaches $I_{max} \pm I_{tol}$ within $T_{rampmax}$. No sample exceeds $I_{max} + I_{tol}$.
5	Hold for T_{stable}	Current stays within $I_{max} \pm I_{tol}$. No fault, no derating event. Meter values are sent periodically.
6	Compare Meter values energy with the power analyzer energy	Deviation within metering tolerance (TBD).
7	EV emulator ends the session	Charging → Finishing. Current goes to 0 A. Contactors open. $V_{out} \leq V_{safe}$.
8	Unplug the connector	Finishing → Idle. The meter is consistent with the measurement.

Pass Criteria

- All expected results are met.
- The current never exceeds $I_{max} + I_{tol}$ at any sample.
- Billed energy matches measured energy within tolerance.
- No transition to Faulted.

Post-conditions

- The charger is in Idle and Available.
- The transaction is closed on the CSMS with correct energy.

Priorities and what's left out

Priorities:

- Safety. Overcurrent leads to cable overheating. TC-02 and TC-04 cover this
- Availability. A customer shouldn't be stranded by a false fault at a legitimate limit. TC-01 and TC-02 cover this
- Billing. Meter consistency at high current. TC-01 and TC-04 cover this

Boundary value analysis gives $I_{max}-1$, I_{max} , and $I_{max}+1$. The large value test catches overflows and saturation bugs. Enforced by TC-03

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Left out on purpose:

Rest	Reason
Thermal soak	Takes hours. Candidate for nightly runs
Real-vehicle tests	Belongs in a separate campaign
Combined faults	Combinatorial explosion. Focus on single failures first.
HMI	Affects UX, less likely to have critical consequences

Exercise 2

Working with AI

AI prompt:

You are a test engineer for DC fast chargers (EVSE)

The charger firmware has a session management state machine with states Idle → Preparing → Charging → Finishing → Faulted

Write a detailed test case to validate the correct sequence for interrupting a charging session started by the user, from the STOP command on the HMI up to reaching a safe state

Include test ID, objective, preconditions, test data, steps, and expected results

Reference relevant standards where applicable

Raw AI draft

Test Case ID: TC-SM-STOP-001

Title: User-initiated charging session stop via HMI

Priority: High

Objective: Verify that when the user presses STOP on the HMI during an active charging session, the charger safely terminates energy transfer, transitions through the correct states, and reaches a safe state in compliance with IEC 61851-23 and OCPP 1.6.

Preconditions

- The charger is powered on and in Idle state.
- The charger is connected to the CSMS via OCPP 1.6J and the connection is stable.
- A CCS2-compatible EV (or EV simulator) is connected.
- The user has been authorized via RFID and a charging session has been started.
- The charger is in Charging state and delivering power (e.g., 400 V, 100 A).
- No active faults are present.

Test Data

Parameter	Value
Connector type	CCS2
Output voltage	400 V DC
Charging current	100 A
RFID tag	Valid test tag (ID: 04A1B2C3D4)
OCPP version	1.6J

Test Steps and Expected Results

Step	Action	Expected Result
1	Verify the charger is in Charging state	HMI displays charging screen with live power, energy and time. Session state = Charging.
2	Press the STOP button on the HMI	HMI displays a “Stopping charging session...” message immediately.
3	Observe state transition	Session state transitions from Charging to Finishing within 1 second.
4	Observe power module behavior	Charging current ramps down to 0 A within 2 seconds, as required by IEC 61851-23.
5	Observe EV communication	Charger sends a stop request to the EV via ISO 15118 / DIN 70121. The EV acknowledges.
6	Observe contactors	DC contactors open once current reaches 0 A.
7	Observe output voltage	Output voltage drops below 60 V DC within 1 second, according to IEC 61851-23 clause 102.5.2.
8	Observe connector lock	Connector is unlocked, allowing the user to remove the cable.
9	Observe OCPP messages	Charger sends StopTransaction.req to the CSMS with reason = “Local” and the correct meterStop value. CSMS responds with StopTransaction.conf.
10	Observe HMI	HMI displays a session summary (energy delivered, duration, cost) and a “Please unplug your vehicle” message.
11	User unplugs the cable	Session state transitions from Finishing to Idle. Charger sends StatusNotification.req with status = “Available”.

Pass Criteria

- All expected results are met.
- The charger reaches a safe state with no residual voltage on the connector.
- No transition to Faulted state occurs.
- The system behaves gracefully and without errors throughout the sequence.

Post-conditions

- The charger is in Idle state and ready for a new session.
- Transaction data is correctly stored and transmitted to the CSMS.

AI draft comments (1/2)

Test Case ID: TC-SM-STOP-001

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Objective: Verify that when the user presses STOP on the HMI during an active charging session, the charger safely terminates energy transfer, transitions through the correct states, and reaches a safe state in compliance with IEC 61851-23 [AI-UNVERIFIED] and OCPP 1.6 [ASSUMP]

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Preconditions

- The charger is powered on and in Idle state
- The charger is connected to the CSMS via OCPP 1.6J [ASSUMP]
- A CCS2-compatible EV (or EV simulator) is connected
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Test Data

Parameter	Value
Connector type	CCS2 [ASSUMP]
Output voltage	400 V DC [ASSUMP]
Charging current	100 A [ASSUMP]
RFID tag	Valid test tag (ID: 04A1B2C3D4) [ASSUMP]
OCPP version	1.6J [ASSUMP]

AI draft comments (2/2)

Test Steps and Expected Results

Step	Action	Expected Result
3	Observe state transition	Session state transitions from Charging to Finishing within 1 second [ASSUMP]
4	Observe power module behavior [ask order confirmation]	Charging current ramps down to 0 A [X A] within 2 seconds [ASSUMP], as required by IEC 61851-23 [AI-UNVERIFIED]
5	Observe EV communication [ask order confirmation]	Charger sends a stop request to the EV via ISO 15118 / DIN 70121 [ASSUMP]. The EV acknowledges [add observation point]
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Pass Criteria

- All expected results are met
- The charger reaches a safe state [ASSUMP] with no residual voltage on the connector
- No transition to Faulted state occurs
- The system behaves gracefully [UNTESTABLE] and without errors throughout the sequence

AI draft comments (2/2)

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Whole draft

- No observation points (i.e., logs)
- Tests only the happy path

Edge cases and missing info

Edge cases:

- STOP during Preparing, CableCheck or PreCharge
- Double STOP
- EV ignores the stop request
- Simultaneous remote stop from the backend
- Backend offline (queue transaction, meter persisted)
- Power loss mid-sequence
- STOP by a non-authorized user

Edge cases and missing info

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- EV ignores the stop request
- Simultaneous remote stop from the backend
- Backend offline (queue transaction, meter persisted)
- Power loss mid-sequence
- STOP by a non-authorized user

Vague or unverifiable in the scenario:

- Define “safe state” measurably
- Determine expected sequence
- Determine end state (Finishing? Idle?)
- Define timing constraints

Pitfalls in using AI blindly

Pitfalls:

- Assumptions → becomes freezing
- Follows code → adapts tests to existing code instead of challenging it
- Hallucinations → specific thresholds, standard clauses and message names that look authoritative
- Confirmation bias → lowered reviewer's scrutiny
- Confidentiality → firmware and logs should not go into non-approved tools

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- Confirmation bias → lowered reviewer's scrutiny
- Confidentiality → firmware and logs should not go into non-approved tools

Benefits:

- Fast prototyping → Get an initial, incomplete draft in seconds
- Reasoning → Use AI as a brainstorming partner and coverage challenger (“what edge cases am I missing?”)
- Boilerplate → Use AI for boring, repeatable tasks and as a formatter
- Checks consistency → Every AI-derived expected result stays tagged [AI-UNVERIFIED] until confirmed
- Repeatability → Keep the prompts in the test repo, so the process is reproducible across releases

Thank you

Q&A

BDR: examples

BDR-001 — Connector locked after EV-side disconnect

ID BDR-001

Observed behavior Unplugging at the vehicle during [Charging](#) drives the session to [Faulted](#). The connector stays locked and the station needs a technician reset; [StopTransaction](#) is sent with `reason = Other`.

FW version 4.8.2 — reproduced on 4.7.0 and 4.8.0

Sources consulted IEC 61851-1, CP state B → A · OCPP 1.6J [StopTransaction.reason](#) enum · developer interview 2026-09-14 · field logs, 11 occurrences across 3 sites

Decision **BUG** — defect raised, fix targeted for 4.9.0

Rationale A customer-initiated disconnect is a normal end of session, not a fault. Locking the cable takes the charge point out of service, so this clears the top rung of the ladder on two counts. The developer confirmed the [Faulted](#) branch was written for CP loss *without* a preceding stop request.

Who decided Test lead + SW owner (session mgmt); safety officer consulted, 2026-09-15

Linked IDs [REQ-SM-014](#) (new) · [TC-SM-201/202/203](#) · defect [EVSE-4471](#)

BDR-002 — Charging continues while the backend is unreachable

ID BDR-002

Observed behavior Loss of the OCPP connection during [Charging](#) does not end the session. Meter values are queued locally and replayed on reconnect; the HMI shows only an offline icon.

FW version 4.8.2

Sources consulted OCPP 1.6J offline-transaction behavior · SW owner interview 2026-09-16 · CPO operations feedback [OPS-882](#) · commit message of the original change

Decision **INTENDED** — undocumented, promoted to a requirement

Rationale Ending the session would strand a paying customer over a fault that is not theirs, and the queue-and-replay is deliberate, not accidental. The behavior was never wrong, only unwritten — so the fix is documentation, not code. Promoted with its limits made explicit rather than left implicit.

Who decided Test lead + SW owner + product owner, 2026-09-16

Linked IDs [REQ-SM-031](#) (new, from this record) · [TC-SM-310/311](#) · [TC-SM-312](#) (queue overflow)

BDR-003 — Silent derate on power module error

ID BDR-003

Observed behavior A single power module fault during [Charging](#) does not end the session: the charger derates to the healthy modules and continues at reduced current. The HMI shows nothing; the customer sees only a slower charge.

FW version 4.9.0-rc1

Sources consulted Developer interview 2026-09-17 — derate deliberate, HMI behavior “not specified” · no written spec found · HMI style guide silent on derate

Decision **PROVISIONAL** — accepted for 4.9.0, expires 2026-12-01

Rationale Derating beats aborting, and that half is intended. Whether the customer must be *told* is a product decision nobody in the room could make, so it is recorded as an open question instead of being frozen by silence. The test pins the behavior against drift but is tagged [provisional](#): it is evidence of what the firmware does, not yet agreement that it is right.

Who decided Test lead; escalated to product owner — open at time of writing

Linked IDs [REQ-SM-047](#) (provisional) · [TC-SM-420](#), tagged [provisional](#) · open question [Q-SM-009](#)

Exercise 1: test cases

Exercise 1: Test Case TC-01

Test Case ID: TC-SM-IMAX-001 (TC-01)

Title: Start charging session with EV requesting exactly the connector maximum current

Priority: High

Objective: Verify that when the EV requests a current equal to I_{max} , the charger goes through Idle → Preparing → Charging, delivers I_{max} within tolerance without overshoot, keeps it stable, and meters the energy correctly.

Preconditions

- Firmware version recorded.
- The charger is powered on and in Idle state.
- The charger is connected to the CSMS simulator.
- An EV emulator is connected.
- A power analyzer is on the DC output.
- I_{max} is configured, and its value is recorded from the charger configuration.
- No active faults are present.

Test Data

Parameter	Value
Output voltage	400 V DC
EV requested current	$I_{max} = 400\text{ A}$
Hold time	T_{stable}

Test Steps and Expected Results

Step	Action	Expected Result
1	EV emulator runs charge parameter discovery with EV max current = I_{max}	Charger advertises EVSE max current $\geq I_{max}$ (checked in the protocol log).
2	Let cable check and precharge complete	Both complete without error. No isolation fault.
3	EV emulator requests target current = I_{max}	State Preparing → Charging.
4	Observe the current ramp-up on the power analyzer	Current reaches $I_{max} \pm I_{tol}$ within $T_{rampmax}$. No sample exceeds $I_{max} + I_{tol}$.
5	Hold for T_{stable}	Current stays within $I_{max} \pm I_{tol}$. No fault, no derating event. Meter values are sent periodically.
6	Compare Meter values energy with the power analyzer energy	Deviation within metering tolerance (TBD).
7	EV emulator ends the session	Charging → Finishing. Current goes to 0 A. Contactors open. $V_{out} \leq V_{safe}$.
8	Unplug the connector	Finishing → Idle. The meter is consistent with the measurement.

Pass Criteria

- All expected results are met.
- The current never exceeds $I_{max} + I_{tol}$ at any sample.
- Billed energy matches measured energy within tolerance.
- No transition to Faulted.

Post-conditions

- The charger is in Idle and Available.
- The transaction is closed on the CSMS with correct energy.

Exercise 1: Test Case TC-02

Test Case ID: TC-SM-IMAX-002 (TC-02)

Title: EV requests current above the connector maximum

Priority: High

Objective: Verify that when the EV requests a current above I_{max} , the charger limits the delivered current to I_{max} without rejecting or faulting the session, and handles extreme request values without overflow.

Preconditions

Same as TC-SM-IMAX-001.

Test Data

[the test is run once per requested value]

Parameter	Value
Connector type	CCS2
EV requested current	Run A: $I_{\text{max}} + 1 \text{ A}$ · Run B: $2 \times I_{\text{max}}$ · Run C: maximum value the protocol allows
Hold time	T_{stable}

Test Steps and Expected Results

Step	Action	Expected Result
1	EV emulator runs charge parameter discovery	Charger advertises EVSE max current = I_{max} .
2	EV emulator requests the value for this run	Session is not rejected. Preparing → Charging.
3	Observe the ramp-up	No sample exceeds $I_{\text{max}} + I_{\text{tol}}$.
4	Hold for T_{stable}	Current stays within $I_{\text{max}} \pm I_{\text{tol}}$. No Faulted state.
5	Run C only: check the logs and the charger’s reaction	No overflow or wrap-around (current does not jump to a low or negative value). No process crash or restart.

Step	Action	Expected Result
6	End the session and unplug	Normal Finishing → Idle. StopTransaction is correct.

Pass Criteria

- The current is limited to $I_{max} + I_{tol}$ in all three runs.
- The session continues. It is neither rejected nor faulted.
- The extreme value (Run C) is handled safely.

Post-conditions

- The charger is in Idle and Available.

Exercise 1: Test Case TC-03

Test Case ID: TC-SM-IMAX-003 (TC-03)

Title: EV requests current just below the connector maximum

Priority: Medium

Objective: Verify that a request just below I_{max} is delivered exactly as requested. This rules out off-by-one errors and overly aggressive limits.

Preconditions

Same as TC-SM-IMAX-001.

Test Data

Parameter	Value
Output voltage	400 V DC
EV requested current	$I_{\text{max}} - 1 \text{ A} = 199 \text{ A}$
Hold time	T_{stable}

Test Steps and Expected Results

Step	Action	Expected Result
1	EV emulator requests $I_{\text{max}} - 1 \text{ A}$	Preparing → Charging.
2	Hold for T_{stable}	Delivered current = requested value $\pm I_{\text{tol}}$. It is not limited lower.
3	End the session and unplug	Normal Finishing → Idle.

Pass Criteria

- The delivered current matches the request within tolerance for the whole T_{stable} period.

Post-conditions

The charger is in Idle and Available.

Exercise 1: Test Case TC-04

Test Case ID: TC-SM-IMAX-004 (TC-04)

Title: Power module fault while charging at the connector maximum current

Priority: High

Objective: Verify that a power module fault while charging at I_{max} brings the charger to Faulted and a safe state within the time limits, closes the session with correct metering, and blocks new sessions until the fault is cleared.

Preconditions

- Same as TC-SM-IMAX-001.
- A fault injection mechanism is available: either a software hook or a hardware fault on the power module.

Test Data

Parameter	Value
Output voltage	400 V DC
EV requested current	$I_{max} = 400\text{ A}$
Hold time	T_{stable}
Injected fault	Power module error

Test Steps and Expected Results

Step	Action	Expected Result
1	Run TC-SM-IMAX-001 steps 1–5	Charging at I_{max} , stable.
2	Inject the power module fault and record the time t0	State Charging → Faulted.
3	Measure the current	Current reaches 0 A within $T_{shutdownmax}$ from t0.
4	Observe the contactors and the output voltage	Contactors open. $V_{out} \leq V_{safe}$ within the specified time (TBD).

Step	Action	Expected Result
5	Observe the OCPP messages	StatusNotification = Faulted with errorCode and vendorErrorCode. StopTransaction.req with a meterStop matching the measurement.
6	Observe the HMI	An error / out-of-service message is displayed.
7	Try to start a new session	Authorization or start is refused while the fault is active.
8	Clear the fault	Faulted → Idle (or reset required, per OQ-05). StatusNotification = Available.

Pass Criteria

- The safe state is reached within the time limits.
- The final billed energy equals the measured energy up to t0.
- No new session is possible while the fault is active.

Post-conditions

- The fault is cleared and the charger is in Idle and Available.